

Technical change and health: Evidence from the reduction of sugarcane burning in Brazil

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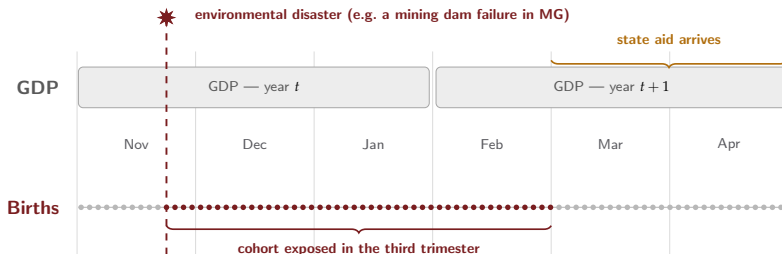
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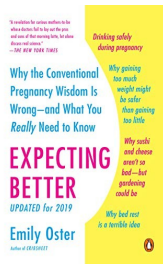
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Health at birth: a thermometer that reads every day



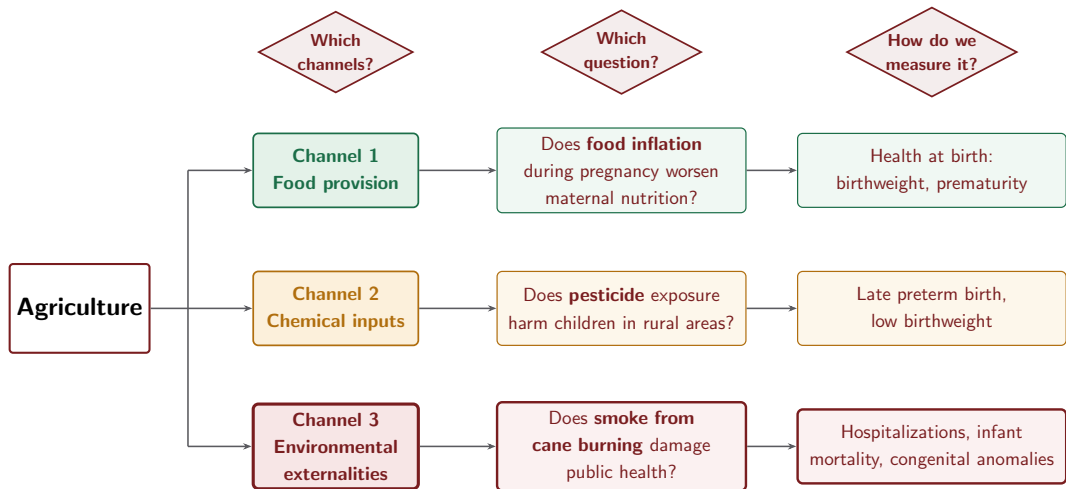
- **GDP** is a great summary of development, but it arrives **once a year and noisy**: a November shock bleeds into December, then into January–February, and by March–April state aid is already mixed into the same figure
- **Babies are born all the time** — about **47 a day in Curitiba** (17,067 in 2024). A third-trimester shock registers at once: birthweight falls, congenital anomalies rise



Emily Oster (Brown),
Expecting Better (2013)

Terrible for the children; excellent for measurement. Birth records are **high-frequency, precisely dated, and cover every birth** — a thermometer of local conditions that annual GDP cannot match.

How does agriculture affect health?

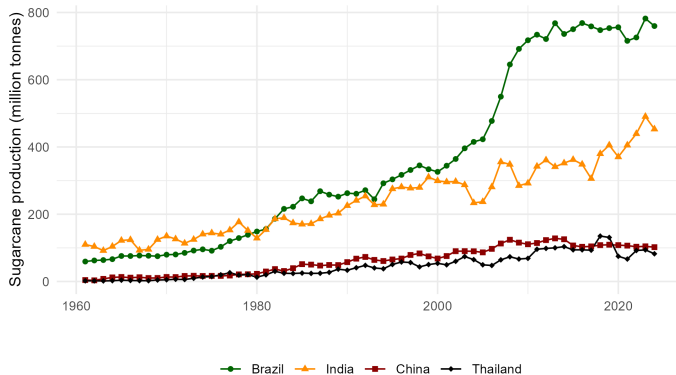


Can a technical change in agriculture improve health?

- Between 2007 and 2015, São Paulo's sugarcane harvest shifted from roughly one-third mechanized to over 95% mechanized, eliminating pre-harvest burning
- We estimate the health effects of this transition using administrative microdata – and the results are striking:

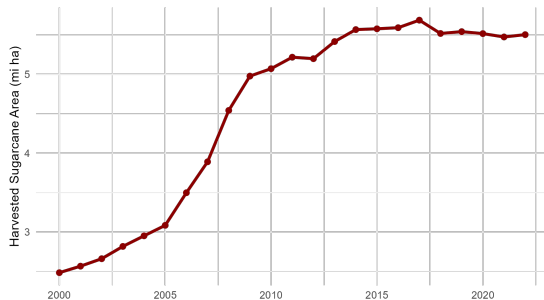
A one-SD increase in exposure reduces respiratory hospitalizations by **7.2%**, circulatory hospitalizations by **4.5%**, cardiac birth malformations by **38%**, and infant respiratory mortality by **11%**. The smoke mechanism is confirmed directly: pollution fell exclusively during harvest months, and municipalities downwind of sugarcane fields benefited most.

Context



- Brazil: **40%** of global sugarcane production (2024). São Paulo alone: **52%** of Brazil's output — roughly **21%** of world production. If São Paulo were a country, its output would nearly match India, the world's #2 producer.

Harvested Sugarcane Area in São Paulo



Harvested area expanded through the late 2000s then stabilized — burning fell while cultivated area held steady, isolating the **technology channel** from the expansion channel.

Where We Sit in the Literature

- Rangel and Vogl [2019]: wind-based smoke exposure → lower birthweight, shorter gestation (São Paulo)
- Chagas et al. [2016]: spatial DiD, sugarcane → respiratory hospitalizations, but sample overlaps expansion *and* burning decline, and treatment is a binary median-split

Our contribution

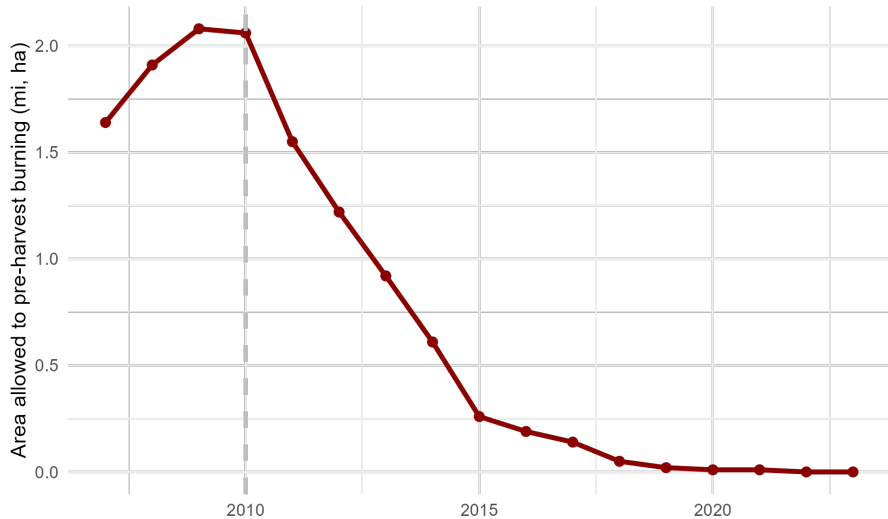
- Broader outcomes: circulatory disease, infant mortality, a biologically-grounded congenital anomaly index
- A sample period that **isolates the burning decline** from the expansion period
- **Continuous** (not binary) exposure, plus three sharper designs (season, wind, dose)

Our Source of Identification

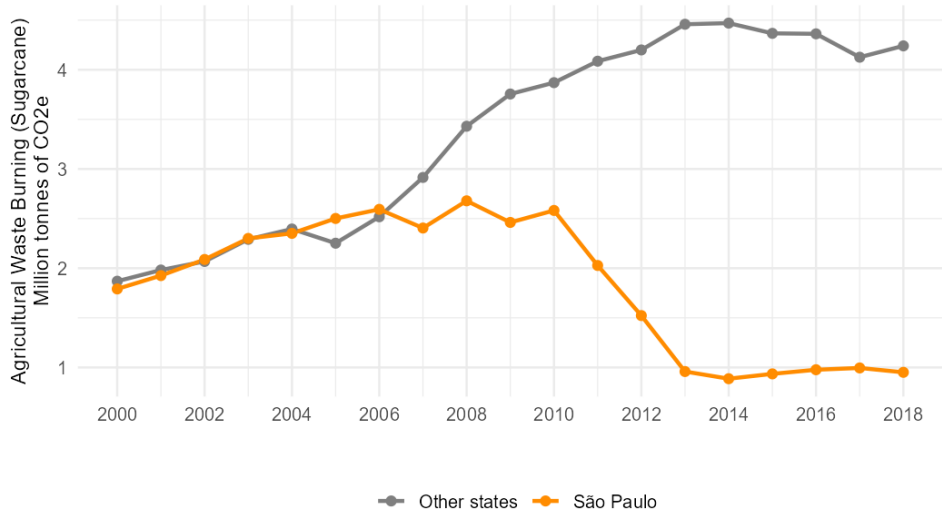
There was a reduction in burning [1]

- **State Law 11.241/2002:** gradual ban on straw burning (targets: 50% by 2011, 100% by 2021)
- **Green Ethanol Protocol (2007):** voluntary industry agreement that accelerated deadlines to 2014–2017 in exchange for preferential financing and export market access
- Rising rural wages and export sustainability standards reinforced the shift [Davis, 2017]

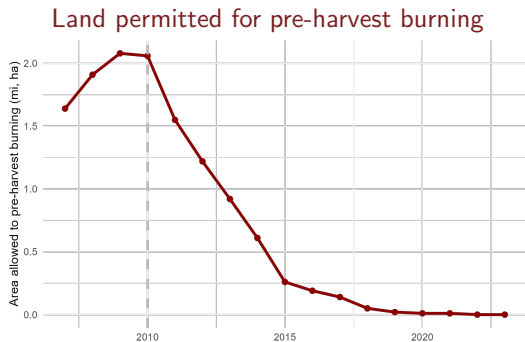
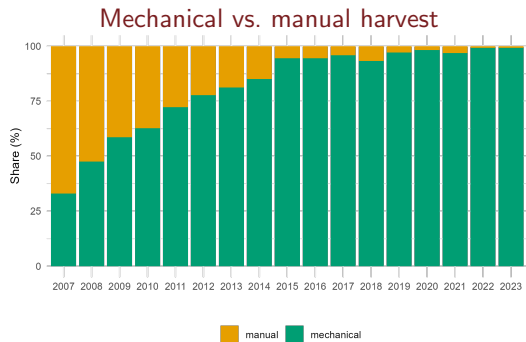
There was a reduction in burning [2]



There was a reduction in burning [3]

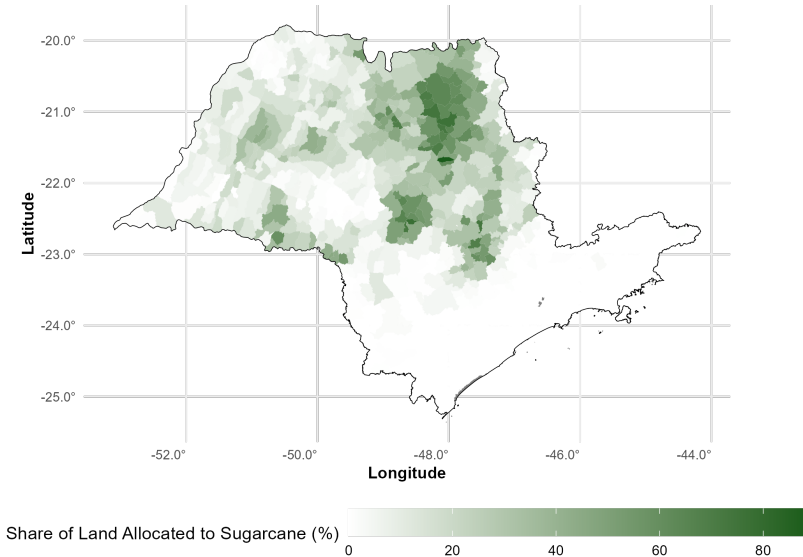


Mechanization and the End of Burning

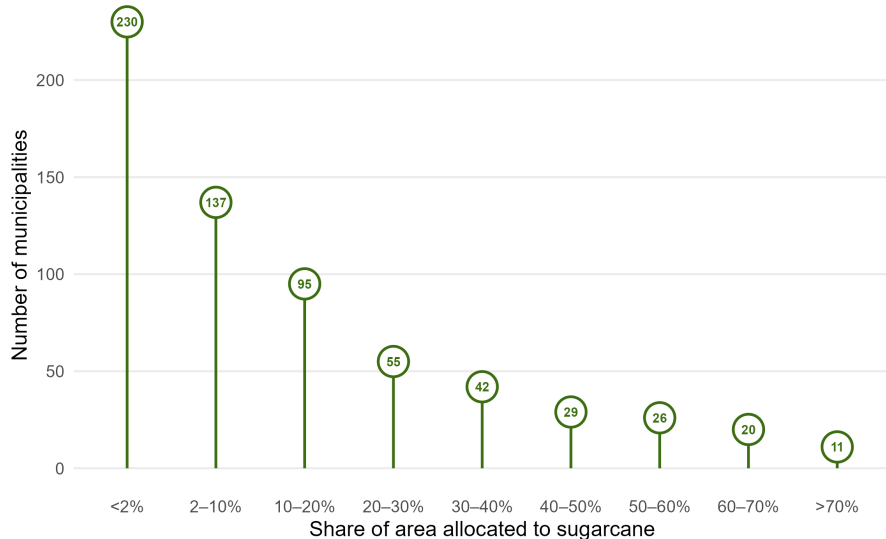


The two margins move together: mechanization is what made the burning ban feasible.

With a spatial heterogeneity in this reduction...



With a spatial heterogeneity in this reduction...[2]



Data and Empirical Strategy

Data Sources

- **Hospitalizations:** SIH/SUS, 2008–2022 (ICD J00–J99, I00–I99)
- **Births:** SINASC, 2007–2023, $N \approx 9.7\text{M}$ individual records
- **Mortality:** SIM, 2002–2023
- **Treatment:** PAM-IBGE sugarcane land share, pre-2011 average
- **PM10/PM25:** CETESB monitoring network, 2003–2024
- **Fire:** MODIS MCD64A1 $\times$ MapBiomas LULC (cane pixels), Google Earth Engine
- **Wind:** NASA POWER/MERRA-2 reanalysis, Jul–Sep 2002–2022

All sources are administrative and cover the universe of events — no survey sampling error.

Summary Statistics

Variable	<i>N</i>	Mean	SD	Max
Sugarcane land share (2002–2010)	642	0.150	0.190	0.886
Respiratory hosp. per 1,000 pop.	8,385	7.281	4.939	45.828
Circulatory hosp. per 1,000 pop.	8,385	6.470	3.085	25.760
Any anomaly (per 1,000 births)	11,882	18.662	91.807	1,000
Infant resp. mortality (per 1,000 births)	9,029	0.702	3.204	111.111

Note the spread in the treatment variable: $SD(Land_i) = 0.19$ against a mean of 0.15 — exposure is highly unequal across the 642 municipalities of São Paulo.

Combining the timing and the spatial heterogeneity

Difference-in-differences, continuous exposure

$$\ln(Y_{it}) = \alpha \cdot TC_{it} + \theta_i + \gamma_t + \epsilon_{it}$$

$$TC_{it} = Land_i \times D_{t>2010}, \quad Land_i = \text{pre-2011 sugarcane land share}$$

- θ_i, γ_t : municipality and year FE; regressions weighted by population (or births); SEs clustered by municipality
- $SD(Land_i) = 0.19$ — the unit for all “one-SD” effects below
- Individual births: analogous LPM with maternal education, sex, marital status FE
- Three sharper designs: harvest-season triple-diff, wind-direction triple-diff, continuous treatment intensity ($fase_t$)

Results

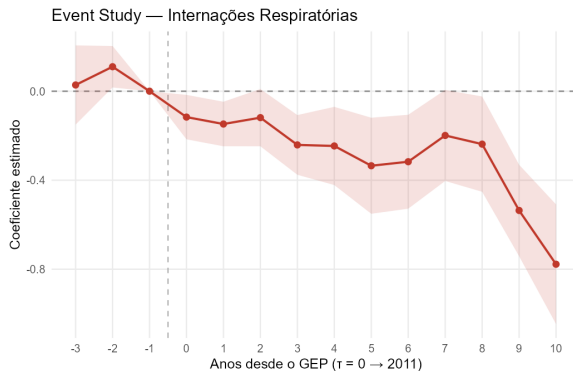
Hospitalizations Fall for Respiratory & Circulatory Disease

$y = \ln(\text{hosp. per capita})$	Full (2008–2021)		Pre-Covid (2008–2019)	
	(1)	(2)	(3)	(4)
<i>Panel A: Respiratory (J00–J99)</i>				
Tech. Change	−0.380*** (0.114)	−0.369*** (0.112)	−0.287** (0.111)	−0.286** (0.111)
<i>Panel B: Circulatory (I00–I99)</i>				
Tech. Change	−0.238*** (0.078)	−0.224*** (0.075)	−0.234*** (0.078)	−0.222*** (0.075)
log(GDP/cap) control		X		X
Municipality & Year FE	X	X	X	X

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Weighted by population, clustered SEs.

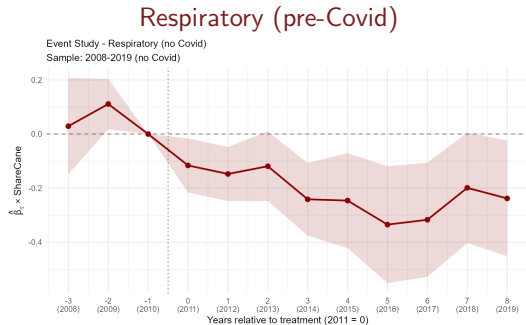
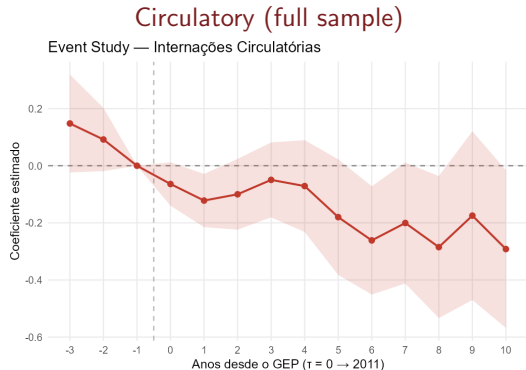
- **1 SD more exposure** $\Rightarrow$ respiratory $\downarrow$ **7.2%**, circulatory $\downarrow$ **4.5%**; both stable when controlling for GDP/capita

Identification: Parallel Trends Hold



Pre-treatment coefficients ($\tau = -3, -2$) near zero and insignificant; post-treatment effects grow steadily as the GEP phases in — consistent with a gradual, accelerating rollout, not a one-time break.

Event Study — Circulatory and Pre-Covid Samples



The same pattern survives dropping the pandemic years, so the result is not a Covid artifact.

Birth Outcomes: No Effect on Weight, Fewer Anomalies

	Any Anomaly (Q00–Q99)	Pol3 Index (NTD+Orofac.+Card.)	LBW (<2,500g)	VLBW (<1,500g)
Tech. Change	−0.006 66* (0.003 90)	−0.004 27** (0.001 82)	0.001 24 (0.002 00)	0.000 10 (0.000 84)
Baseline mean	11.45/1000	3.59/1000	9.26%	1.47%

Individual (Sinasc), $N = +9M$, Controls: Mun., year, educ., sex, marital, race...

- **No effect on birthweight** (LBW, VLBW)
- **Congenital anomalies fall 11%**; the **Pol3 pollution-linked index falls 22.6%** at 1 SD of exposure
- Pol3 = neural tube defects + orofacial clefts + cardiac malformations

Which Anomalies? The Biology Lines Up

$y = \text{Pr}(\text{anomaly type})$	Neural Tube (Q00–Q07)	Orofacial (Q35–Q37)	Cardiac (Q20–Q26)	Pol3 Index
Tech. Change	−0.000 539** (0.000 221)	0.000 000 (0.000 156)	−0.003 846** (0.001 623)	−0.004 265** (0.001 823)
Baseline (per 1,000)	1.09	0.73	1.91	3.59
Relative effect (1 SD)	−9.4%	≈ 0	−38.3%	−22.6%

- **Cardiac malformations** drive the effect: fine PM crosses the placental barrier, disrupts conotruncal septation
- **Orofacial clefts: null** — their teratogenic pathway runs through pesticides, not combustion PM; cane is not pesticide-intensive
- No socioeconomic gradient: effects similar across maternal education groups

Mortality: A Dose-Response Gradient

Adults — no effect

	Respiratory	Circulatory
Exp	-0.028 (0.093)	-0.043 (0.071)

⇒ -0.5% and -0.8% at 1 SD —
statistically zero

Infants (<5) — significant

	Resp. mortality
Exp	-0.593* (0.344)

⇒ -11% at 1 SD; all-cause infant mortality:
null

Infants are physiologically more susceptible. PM reductions are large enough to move infant respiratory mortality even though they fall short of the threshold for adult mortality.

Mechanism: Is It Really the Smoke?

Mechanism: Less Fire → Less Particulate Matter

	Harvest (Jul–Sep)	Off-season (Jan–Mar)
<i>Panel A: Sugarcane burning (MODIS × MapBiomas)</i>		
ln(burned area, cane pixels)	−4.964*** (0.450)	—
<i>Panel B: Particulate matter (CETESB)</i>		
ln(PM10)	−0.094*** (0.010)	−0.005 (0.006)
ln(PM25)	−0.091*** (0.010)	−0.001 (0.006)

*** $p < 0.01$. Off-season burned area near-zero throughout (placebo).

- At mean exposure, harvest-season burned area on cane fields falls $\approx$ **66%**
- PM10 ↓1.8%, PM25 ↓1.7% at 1 SD — **only during harvest**; off-season effects are null

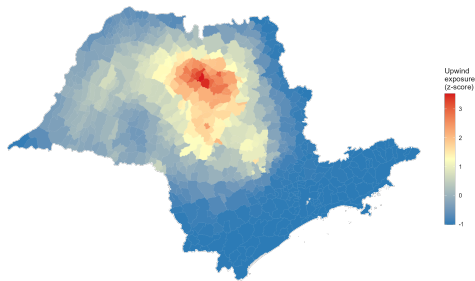
Mechanism: Health Gains Concentrate in Harvest Months

$$\ln Y_{itm} = \alpha \cdot Exp_{it} + \beta \cdot Exp_{it} \times Harvest_m + \theta_{i \times m} + \gamma_t + \varepsilon_{itm}$$

$y = \ln(\text{hosp.}/\text{pop})$	Respiratory	Circulatory
Exp (off-season)	-0.135*** (0.033)	-0.083** (0.028)
$Exp \times Harvest$ (Jul-Sep)	-0.069*** (0.011)	-0.014 (0.013)

Respiratory hospitalizations fall *exactly* when burning would have occurred. Circulatory shows no such spike, consistent with a more diffuse pathway. **Three independent data sources (fire, PM, hospitalizations) all point to the smoke channel.**

Mechanism: Smoke Travels with the Wind



	Respiratory		Circulatory	
	Full	Pre-Cov.	Full	Pre-Cov.
Exp	−0.348***	−0.265***	−0.238***	−0.222***
WindExp	−0.035*	−0.027	−0.033	−0.011

Municipalities **downwind** of cane fields during the burning season gain more than upwind ones with identical own-exposure. Wind direction is plausibly exogenous to municipal health trends.

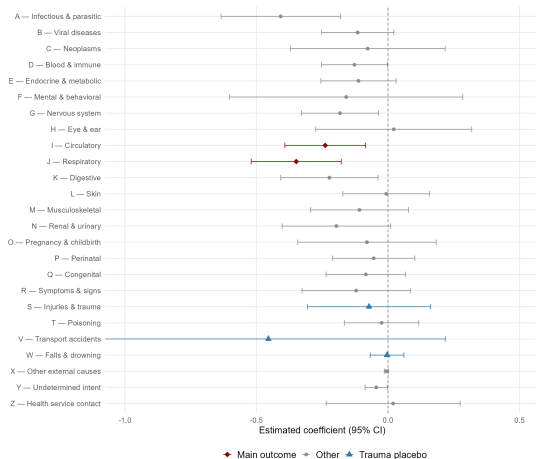
Robustness

Four Independent Checks

- **Wind direction:** downwind municipalities benefit more ($WindExp = -0.035^*$, respiratory) — smoke travels with the wind
- **Placebo across all ICD-10 chapters:** trauma, transport accidents, falls (no biological link to PM) are null; effects concentrate in respiratory and circulatory chapters
- **Continuous treatment** ($fase_t$, fraction of emissions already eliminated): implied effect at mean $fase_t \approx 0.56$ is -0.346 — nearly identical to the binary estimate (-0.344)
- **Excluding São Paulo city** (12M pop., zero cane share): effects survive, -0.327^{**} (resp.) vs. -0.380^{***} baseline

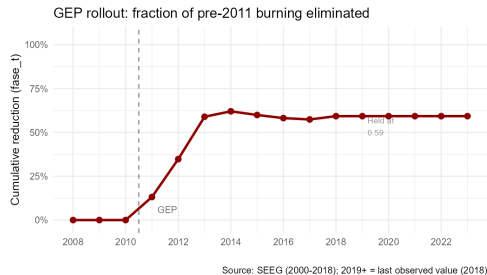
The next three slides show each one.

Placebo Across All ICD-10 Chapters



If the design were picking up general improvements in municipal health infrastructure, every chapter would move. It does not: only the two chapters with a biological link to particulate matter respond.

Continuous Treatment Intensity



	Respiratory	Circulatory
Binary (0/1)	−0.344***	−0.239***
Continuous ($\times fase_t$)	−0.618***	−0.393**

Replacing the post-2010 dummy with $fase_t$ — the fraction of burning emissions already eliminated — gives an implied effect at the mean of -0.346 , essentially the binary estimate. The result is not an artifact of where we cut the treatment date.

Robustness: Excluding São Paulo City

	Respiratory		Circulatory	
	Full	Pre-Covid	Full	Pre-Covid
Baseline	-0.380***	-0.287**	-0.238***	-0.234***
Excl. SP city	-0.327**	-0.236*	-0.228**	-0.189**

São Paulo city is 12M people with zero cane share — a heavily weighted, mechanically untreated observation. Dropping it leaves the estimates essentially unchanged.

What Are the Stakes?

- Aggregating the hospitalization reduction over 2011–2021 across exposed municipalities: **≈118,000** fewer hospital admissions (71,000 respiratory + 47,000 circulatory)
- At 2019 SUS reimbursement rates: **R\$ 209 million** (USD 53M) in avoided hospitalization costs
- **≈1,640** fewer births with cardiac malformations (2011–2023) $\Rightarrow$ **R\$ 22 million** in avoided surgical costs

R\$ 232 million (USD 59 million) in avoided costs over a decade — a conservative *lower bound* that excludes outpatient care, productivity losses, lifetime disability transfers, and the value of infant lives saved.

Takeaway

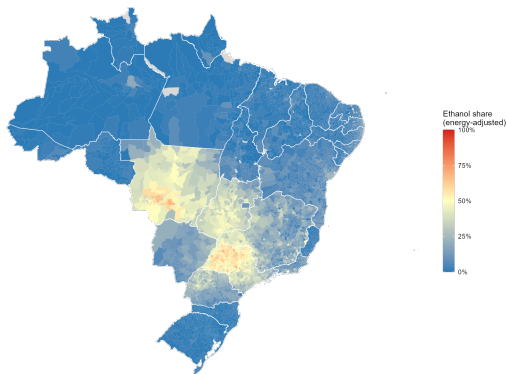
- A gradual, phased ban on pre-harvest sugarcane burning — backed by regulation *and* voluntary export-market incentives — delivered large, well-identified health gains
- The pattern across outcomes traces a **dose-response gradient**: null adult mortality → significant infant mortality → significant hospitalizations → significant congenital anomalies, each margin defined by a different vulnerability threshold
- Relevant beyond São Paulo: Northeast Brazil (Pernambuco: >90% manual), India, Thailand, Florida still rely on pre-harvest burning

Agricultural fire is a widespread, under-regulated externality. Aligning private incentives with public health — as the Green Ethanol Protocol did — can generate large, measurable social returns.

Thank you! Questions?

Backup Slides

Backup: Ethanol Market Context



Energy-adjusted hydrated ethanol share in the fuel mix, 2020–2024. Hydrated ethanol competes directly with gasoline where transport costs from the cane belt are low.

Backup: Aggregate Birth Outcomes (Municipality-Year)

	$\ln(\text{birthweight})$	$\ln(\text{LBW})$	$\ln(\text{VLBW})$	$\ln(\text{anomalies})$
Exp	-0.002 (0.003)	0.027 (0.023)	-0.067 (0.056)	-0.795** (0.311)

Aggregate spec confirms the anomaly result; individual-level LPM (main text) additionally decomposes it by diagnostic group.

References I

- André LS Chagas, Carlos R Azzoni, and Alexandre N Almeida. A spatial difference-in-differences analysis of the impact of sugarcane production on respiratory diseases. *Regional Science and Urban Economics*, 59:24–36, 2016.
- C Austin Davis. Why did sugarcane growers suddenly adopt existing technology? Technical report, Working Paper - Yale Department of Economics, 2017.
- Marcos A Rangel and Tom S Vogl. Agricultural fires and health at birth. *Review of Economics and Statistics*, 101(4):616–630, 2019.